

EXACT-2026 COMPETITION: SOLUTION DESCRIPTION

Team: HuuHiep - Physics & Logic Reasoning Pipeline on Modal

1. Datasets Used

Our solution utilizes a combination of the official EXACT-2026 datasets and synthesized data to train and fine-tune the pipeline components:

- **Dataset 1 (Logic Deduction):** Consists of ~2,000 logic queries requiring first-order logic (FOL) deduction from set premises. We used the official train/validation sets to fine-tune the Logic LoRA adapter. Example query: 'If all cats are animals, and Fluffy is a cat, is Fluffy an animal?' -> Answer: 'Yes'.
- **Dataset 2 (Physics Calculation):** Consists of ~540 physics word problems covering chapters like electrostatics, capacitance, and circuit analysis. We augmented this dataset with ~1,200 synthetic physics problems generated by matching formula templates to fine-tune the Physics LoRA adapter. Example query: 'Calculate the energy stored in capacitor C when $C = 100 \mu\text{F}$ and $U = 30 \text{ V}$.' -> Answer: '0.045 J'.

2. Technical Approach and Methodology

We implemented a robust hybrid architecture tailored separately for Logic and Physics tasks, deployed on Modal:

- **Type 1 (Logic Pipeline):** Routes queries to the fine-tuned 'exact-lora' adapter. The model extracts the minimum relevant premises, performs step-by-step logical deduction, and outputs the answer matching the multiple-choice options.
- **Type 2 (Physics Pipeline):** Combines three layers of validation to achieve maximum accuracy:
 - Retrieval-Augmented Generation (RAG): Uses Qdrant Cloud to search and retrieve physical laws and formulas corresponding to the question's topic.
 - Program-Aided Language (PAL) Sandbox: The fine-tuned 'exact-lora-type2' model converts the natural language question into executable Python code. The code is executed in a secure sandbox to calculate the exact numeric value and unit, eliminating LLM math hallucinations.
 - Deterministic Solver: Operates as a verification layer. It checks if the problem matches known physical topologies (like capacitor networks or vector forces) and computes a verified answer deterministically.

3. Model Size and Parameter Constraints Verification

To ensure strict compliance with the competition's constraint that the total parameters of all loaded/active LLMs must not exceed 8B at any given time, our pipeline uses a sequential adapter loading strategy on Modal:

Task / Route	Base Model	LoRA Adapter	Active Parameters
Type 1 (Logic)	Qwen 2.5 7B / Qwen 3 8B	exact-lora	~8.0 Billion
Type 2 (Physics)	Qwen 2.5 7B / Qwen 3 8B	exact-lora-type2	~8.0 Billion
Static Solvers	None (Rule-based Python)	None	0.0 (Non-LLM)

Proof of Compliance: The base model served is Qwen3-8B. The LoRA adapters ('exact-lora' and 'exact-lora-type2') are dynamically routed and loaded by vLLM at runtime. Since vLLM loads only one adapter per request and executes requests sequentially, the active parameter size never exceeds the 8.0 Billion threshold at any point. Non-LLM modules (Deterministic Solver, RAG parser) do not contain model parameters and thus are excluded from the limit per Section 6.3 of the guide.